WANKR WAR: Onchain Tournament Mechanics (Unrevealed Era)

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Abstract

This document formalizes the mechanics of the *WANKR WAR* tournament during the unrevealed NFT phase. Entry is onchain (USDC with unique fee amount), eligibility requires holding WANKR NFTs at snapshot, and the prize pool is matched by the DAO to bootstrap excitement. Outcomes are randomized but weighted by holdings, using a transparent Bradley–Terry model. A dual-announcer commentary system provides live entertainment during streams. This paper defines the math, data flow, prize splits, and anti-abuse constraints.

1 Design Goals

- Social-first experience: announcements and results posted via an agent account on X.
- Onchain eligibility: payments + NFT holdings validated via Dune queries or a node API.
- Transparent math: verifiable scoring, randomization, and posted RNG seeds.
- Scalable to $\leq$50 entrants; optimized for quick tournaments and streamable play-by-play.

2 Enrollment Flow

1. **Announcement:** The agent posts entry rules, unique fee (e.g. 13.37 USDC), window $[t_0, t_1)$, and requirement to send from the wallet holding WANKR NFTs.
2. **Payment capture:** All transfers are queried via Dune SQL:

   `token = USDC, to = treasury, value = fee_units, ts` $\in$ `[t_0, t_1)`.

3. **Eligibility check:** For each payer wallet, snapshot WANKR NFT balances at block b_{snap}. If $\sum_c n_c > 0$, entrant is eligible; otherwise added to refund ledger.
4. **Snapshot lock:** Entrant balances $\{n_c\}$ are frozen at b_{snap}.

3 Scoring

Let n_c = count in category c and K_c = effective cap (e.g. 10 or total minted that day). The per-category factor is:

$$f_c(n_c, K_c) = \frac{\log(1 + n_c)}{\log(1 + K_c)} \in [0, 1]. \tag{1}$$

Weighted war score:

$$W_{\text{raw}} = \frac{\sum_c w_c f_c}{\sum_c w_c}, \quad S = \max(S_{\min}, 100 \cdot W_{\text{raw}}). \tag{2}$$

Default $S_{\min} = 10$. This ensures diminishing returns, while rewarding breadth.

4 Match Resolution

Two players A, B with skills S_A, S_B have outcome probability:

$$P(A \text{ wins}) = \frac{S_A}{S_A + S_B}. \tag{3}$$

Draw uniform $U \sim \mathcal{U}(0,1)$. If $U < P(A)$ then A wins, else B. Seed string (event, match, block hash) is posted for auditability.

5 Prize Pool

If N entrants each pay f (e.g. 13.37 USDC), the entrant pool is:

$$\text{pool}_{\text{in}} = N \cdot f. \tag{4}$$

The DAO match is $\text{pool}_{\text{dao}} = \text{pool}_{\text{in}}$. Total prize pool:

$$\text{prize} = \text{pool}_{\text{in}} + \text{pool}_{\text{dao}}.$$

Default split: 50/30/20 for 1st/2nd/3rd. Example: $N = 50, f = \$13.37 \implies \text{pool}_{\text{in}} \approx \$668.5$, matched to \$1337, prizes: \$668, \$401, \$268.

6 Tournament Lifecycle

1. Announce entry rules and stream time.
2. Enrollment window open (24h).
3. Close enrollment; lock snapshots; post bracket image.
4. Run matches: agent posts pre-roll & result for each bracket match.
5. Stream commentary with dual announcers (Eleven Labs AI voices).
6. Podium post: image with 1st/2nd/3rd, winnings, and NFT badges.

7 Commentary System

- **Announcer A:** factual play-by-play, issues questions.
- **Announcer B:** witty banter, answers questions.
- **Templates:** short lines ($\leq$ 6s audio), pre-cached where possible.
- **Pipeline:** match event $\rightarrow$ template $\rightarrow$ Eleven Labs TTS $\rightarrow$ mixer $\rightarrow$ OBS $\rightarrow$ Twitch/YouTube.

8 Security & Constraints

- Entry must be sent from wallet holding WANKR NFTs.
- Unique fee value avoids collisions with unrelated transfers.
- Snapshot block prevents mid-event transfers from changing outcomes.
- RNG seed published for reproducibility.

9 Future Extensions

- Elo ratings across tournaments.
- Onchain badge mints (Upset, Win Streak, etc.).
- Automated refund transactions via DAO script.
- Cross-chain expansions or cohort-based leagues.